

TUMMALAPALLI SHANMUKHA NAGA SAI

Software Engineer | Java · C/C++ · DSA · OOP · SDLC

+91 9381255060 | shanmukhnagasai21@gmail.com | LinkedIn | GitHub | Hyderabad, Telangana

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science graduate (CGPA 8.49) with strong Systems Software Programming fundamentals in Java and C/C++, backed by a deep grounding in Object-Oriented Design, Operating Systems, Data Structures & Algorithms, and complexity analysis — 500+ problems solved across arrays, graphs, dynamic programming, trees, tries, and heaps. Hands-on experience across the SDLC: designing, building, testing, and refactoring full-stack applications in Java/Spring Boot, delivered in Agile sprints with code review, CI, and clear documentation practices. Comfortable taking ownership of features end-to-end and using discretion to seek help when needed, while collaborating closely with teammates.

TECHNICAL SKILLS

Systems Programming: Java (OOP, Multithreading, Collections, Exception Handling), C++ (fundamentals), C

CS Fundamentals: Data Structures & Algorithms, complexity analysis, Object-Oriented Design, Operating Systems, Computer Networks — 500+ problems solved

SDLC & Practices: Agile/Scrum, CI, code review, refactoring, unit/integration testing, documentation practices

Frameworks & Tools: Spring Boot, Spring Data JPA, REST APIs, Git/GitHub, Docker, Maven, Postman

Databases: MySQL, PostgreSQL, MongoDB, Redis — schema design, query optimization

ENGINEERING PROJECTS

DevBoard – Scalable Sprint & Task Management Platform | [GitHub](#)

2025

- Owned end-to-end design and delivery of a full-stack platform in Java and Spring Boot (MySQL, Redis, React.js) supporting 100+ concurrent users, applying OOP design patterns and clean, modular service-layer architecture.
- Diagnosed a concurrent-load latency bottleneck via systematic profiling and refactoring; implemented Redis caching, cutting API latency by 95% while maintaining correctness across test scenarios.
- Implemented JWT authentication, role-based access control, and unit-tested service components, delivered in Agile sprints.

Idea Evolution Tracker – Full-Stack Data Platform | [GitHub](#)

2025

- Designed and developed a Spring Boot and React application using Spring Data JPA for ORM-based data access, following modular software architecture and SDLC best practices.
- Refactored and optimized backend performance by 70% through SQL query tuning, indexing, and pagination; implemented JWT-secured REST APIs with full documentation.

Trimly – High-Throughput URL Shortener Microservice | [GitHub](#) · [Live on Render](#)

2025

- Built and deployed a REST API (Node.js/Express.js) with MongoDB and Redis caching, improving P99 response time by 60% through systematic testing and backend optimization.

APPLIED RESEARCH EXPERIENCE

Research Author — VIT-AP University Sep 2025 – May 2026

- Built and evaluated deep learning pipelines in Python for medical image classification (AUC 0.97, F1 94%) and multimodal deepfake detection (95% AUC), with technical documentation aligned to reproducible research standards.

EDUCATION

B.Tech in Computer Science & Engineering — Vellore Institute of Technology (VIT-AP)

2022 – 2026

CGPA: 8.49 | Core Coursework: DSA, OOP, Operating Systems, Computer Networks, DBMS, Algorithm Design, Distributed Systems, Software Engineering, AI/ML

XII (BIEAP) — Sri Chaitanya Junior College | Percentage: 81.1%

2020 – 2022

ACHIEVEMENTS & CERTIFICATIONS

- Solved 500+ LeetCode problems spanning arrays, graphs, dynamic programming, trees, tries, heaps, and system design — active weekly competitive programming with strong complexity reasoning.
- Top 20 Finalist — CS-Quest Hackathon, VIT-AP: system design, backend implementation, and solution delivery under competitive time pressure.
- Data Structures & Algorithms with Java (Certified) | Full Stack Development with MERN Stack (Ethnus Certified) | HackerRank SQL (Verified).